

Smart Expense Test Plan

Smart Expense and Budget Management Application

1 Test plan identifier

SEB-TP-001 | Version 1.1 | 17 September 2026

Software Engineering Mini Project | Section 5G | Team 14

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2 Introduction

This plan describes the tests for Smart Expense before the mini project review. We will check that users can manage their expenses and budgets, that the totals are correct, and that each account keeps its data private.

References: Team 14 SRS, version 1.0, dated 6 September 2026; Mini Project Deliverables Part 1; IEEE 829 test plan content.

Repository: <https://github.com/gsathvik73-afk/se-project>

3 Test items

The browser screens, login and session handling, transaction and category functions, budgets, dashboard, CSV export, database and HTTPS setup will be tested. Record the tested build, database and browser versions with the results.

4 Features to be tested

Registration, login, logout and account isolation; adding, editing and deleting income and expenses; input validation; categories; filters and sorting; monthly and category budgets; alerts; dashboard totals and charts; CSV export. We will also check speed, concurrent users, saved data, ease of use, browser support and security.

5 Test approach

Run the tests using sample accounts and a separate database. Start with login and one expense entry, then follow the cases in Section 16. Try both valid and invalid inputs. Compare totals with hand calculations. After a fix, repeat the failed test and any related checks.

5.1 Security validation

For confidentiality, check HTTPS, password hashes and separation of user data. For integrity, check owner permissions, input handling and login controls. TC-SEC-01 to TC-SEC-05 cover all five SRS security requirements. Use two accounts to try accessing and changing another user's records.

6 Features not to be tested

Bank or UPI integration, statement import, shared budgets, native mobile apps, multiple currencies, tax calculations and email or push notifications are outside the SRS scope. Payment mode fields, mobile browser layout and in-app alerts are included.

7 Pass and fail criteria

A test passes when all its expected results are met. Record incorrect behaviour as Fail and an unavailable test environment as Blocked. All tests must be run, all High priority requirements must pass, and no Critical or Major defect may remain at final review.

Critical: data loss or another user's data exposed. Major: a core feature fails or a financial total is wrong. Minor: a display issue that does not stop use. Record remaining Medium priority failures for review.

8 Suspension and resumption

Stop affected tests if the server fails, login stops working or test data is corrupted. Resume after the issue is fixed, the data is restored and login and expense entry work again.

9 Test deliverables

Keep this plan, the test results, a defect list, screenshots or timing records, and a final test summary. Results should include case ID, tested build, tester, date, actual result, status and defect ID. All cases are initially Not run.

10 Testing tasks

Set up the application, create users and sample data, run the tests, record failures, verify fixes and complete the final summary.

11 Test environment

Use the project's application server and database, current Chrome, Firefox and Edge, and a 360 px wide mobile view. Browser tools are enough for normal checks; use a basic load tool for concurrent users. HTTPS checks need a deployed test URL with a valid certificate.

12 Responsibilities and training

Proposed assignments are shown below. Before testing, the team will review the SRS, sample data and result sheet together.

Member	Tests
Sahana Shivakumar Badiger	Usability, browser checks and test results
Sai Arjun M	Login, access control and security
Sai Arya A L	Transactions, categories and saved data
Sai Sathvik Gullipalli	Budgets, dashboard, export and performance

13 Schedule

18 September: setup. 19 to 24 September: testing. 25 to 26 September: fixes and retesting. 27 September: final review. Submission deadline: 28 September 2026.

14 Risks and contingencies

If a feature is unfinished, record the missing requirement and test it in the next build. Back up sample data before restart checks. If hosting fails, run local checks and complete HTTPS checks when hosting is available.

15 Approvals

Team 14 reviewer: _____ Date: _____
Course coordinator: _____ Date: _____

16 Test cases

16.1 Functional tests

Use separate users A and B. Before each independent test, reset A to income INR 10000 and expenses Food INR 500, Travel INR 300 and Rent INR 1000 in the selected current month. Set monthly budget INR 5000 and Food budget INR 1000. Baseline expense is INR 1800 and balance INR 8200. Log in as A unless the test says otherwise.

Case ID	Steps and input	Expected result
TC-AUTH-01	Register with a new email and an 8 character or longer password. Repeat the email.	One account is created; duplicate email is rejected.
TC-AUTH-02	Log in with correct credentials, then try a wrong password.	Correct login opens Dashboard; wrong login shows an error.
TC-AUTH-03	Log out, then open Dashboard directly and retry the old session.	Login is required; the old session cannot access protected data.
TC-AUTH-04	As A, check lists, charts and CSV, then request B's record ID.	Only A's data appears. B's record returns 403 or 404.
TC-EXP-01	Add INR 250, Food, today, UPI, note Lunch. Also try no note.	Values save correctly; with one entry, expense is INR 2050.
TC-EXP-02	Add income INR 2000, today, source Freelance.	Income is INR 12000; balance is INR 10200.
TC-EXP-03	Try 0, -1, blank or text amounts and a future date. Then try 0.01 today.	Invalid input gets a field error and is not saved; valid input saves.
TC-EXP-04	Change the INR 500 expense to 600; reload. Separately edit an income.	Edits persist and totals update; expense edit gives INR 1900.
TC-EXP-05	Delete the INR 300 expense: first cancel, then confirm.	Cancel keeps it. Confirm removes it; expense becomes INR 1500.
TC-EXP-06	Filter by date, category and amount. Sort both ways; try no matches.	Rows and filtered total match; sorting is correct; empty total is zero.
TC-CAT-01	Check the eight SRS defaults. Add Books, use it, then rename it Study.	Defaults and new category are selectable; rename persists.
TC-BUD-01	Set monthly budget INR 5000 and Food budget INR 1000; reload.	Both budgets persist for the selected month.
TC-BUD-02	Compare monthly and Food spending with their saved budgets.	Monthly: 1800 spent, 3200 left, 36%. Food: 500, 500, 50%; within INR 1.
TC-BUD-03	Test below 80%, at 80%, at 100% and above 100%; add another expense.	Warning starts at 80%; distinct alert above 100%; entry stays allowed.
TC-DSH-01	Compare monthly cards with records; switch to an empty month.	Income 10000, expense 1800, balance 8200; empty month has zero totals.
TC-DSH-02	Check category totals and a six month trend with empty months.	Food 500, Travel 300, Rent 1000; trend matches records; empty months are zero.
TC-RPT-01	Filter the list and export CSV. Compare data rows with the screen.	CSV downloads with correct type; rows and values match the filtered list.

16.2 Non functional tests

Use the same tested build. Keep timing records, restart results and user observations with the test results.

Case ID	Steps and input	Expected result
TC-PERF-01	Seed 5000 transactions. Time 100 loads each of Dashboard and Transactions.	At least 90 of 100 loads per page complete within 3 seconds.
TC-PERF-02	Run 20 separate logged in users together to view pages and save entries.	All actions work; no server 5xx errors or mixed user data.
TC-REL-01	Save data and restart the server. Separately force a save to fail before commit.	Saved data survives; failed save leaves no partial record and allows retry.
TC-UX-01	Ask five first-time users to add an expense from Dashboard and correct an invalid amount.	Each finishes within 3 clicks; errors explain how to correct the input.
TC-PORT-01	Check main screens and actions in Chrome, Firefox and Edge, and at 360 px width.	Screens and actions work; text and controls are not clipped.

16.3 Security tests

Run these checks on the test deployment with sample accounts. These cases cover Section 5.1.

Case ID	Steps and input	Expected result
TC-SEC-01	Open HTTP and HTTPS URLs; check certificate, TLS version and session cookie.	HTTP redirects to valid HTTPS with TLS 1.2 or later; cookie is HttpOnly and Secure.
TC-SEC-02	Create two accounts with the same password. Check stored hashes and logs.	Salted bcrypt or PBKDF2 hashes differ; no plain or reversible passwords are stored or logged.
TC-SEC-03	As A, try reading, editing and deleting B's transactions, budgets and categories.	403 or 404 is returned; B's data is unchanged.
TC-SEC-04	Try SQL injection in login and text fields; save a script tag in a note. Check query handling.	No login bypass or query manipulation; script renders as text. Queries use parameters or an ORM.
TC-SEC-05	Try 7 and 8 character passwords. Fail login five times; try again immediately and after 5 minutes.	7 is rejected, 8 accepted. After five failures, login is blocked for 5 minutes, then correct login works.

17 Requirements traceability matrix

Each SRS requirement is linked to its test case below. Use the case ID to connect the test result and any defect to the requirement.

SRS requirement	Requirement	Test case
SEB-F-001	Registration	TC-AUTH-01
SEB-F-002	Login	TC-AUTH-02
SEB-F-003	Logout and protected pages	TC-AUTH-03
SEB-F-004	User data isolation	TC-AUTH-04
SEB-F-005	Add expense	TC-EXP-01
SEB-F-006	Add income	TC-EXP-02
SEB-F-007	Amount and date validation	TC-EXP-03
SEB-F-008	Edit transaction	TC-EXP-04
SEB-F-009	Delete confirmation	TC-EXP-05
SEB-F-010	Filter and sort transactions	TC-EXP-06
SEB-F-011	Default and custom categories	TC-CAT-01
SEB-F-012	Monthly and category budgets	TC-BUD-01
SEB-F-013	Budget usage figures	TC-BUD-02
SEB-F-014	Budget threshold alerts	TC-BUD-03
SEB-F-015	Monthly dashboard totals	TC-DSH-01
SEB-F-016	Category and six month charts	TC-DSH-02
SEB-F-017	Filtered CSV export	TC-RPT-01
SEB-NF-001	Page load with 5000 transactions	TC-PERF-01
SEB-NF-002	20 concurrent logged in users	TC-PERF-02
SEB-NF-003	Durability and rollback	TC-REL-01
SEB-NF-004	Three click entry and errors	TC-UX-01
SEB-NF-005	Browser and mobile compatibility	TC-PORT-01
SEB-SR-001	HTTPS and TLS	TC-SEC-01
SEB-SR-002	Salted password hashes	TC-SEC-02
SEB-SR-003	Owner authorisation	TC-SEC-03
SEB-SR-004	SQL injection and stored XSS	TC-SEC-04
SEB-SR-005	Password length and login lockout	TC-SEC-05